Extra credit: $\frac{1.5}{2}$

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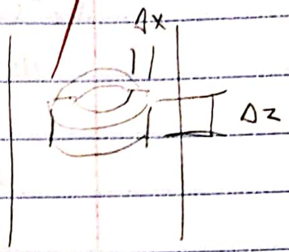
CHE 312
9-26-19

CHE312 HW04

Q1,3 on credit $18/25$

1. For laminar flow through a tube



$$v_z = \frac{(\mathcal{P}_0 - \mathcal{P}_L)R^2}{4\mu L}\left[1-\left(\frac{r}{R}\right)^2\right] = v_{z,max}\left[1-\left(\frac{r}{R}\right)^2\right]$$

Assumptions

1. well established, laminar flow ($v_r = v_\theta = 0$)
2. energy flow is only in z and r direction (no θ)
3. no radiation, nuclear, or other energy transport except conduction from wall, convection due to flow and work.

Shell balance

energy in from r-direction: $(2\pi r e_r \Delta z)|_r$

energy out from r direction: $(2\pi r e_r \Delta z)|_{r+\Delta r}$

energy in from z-direction: $(2\pi r \Delta r e_z)|_z$

energy out from z-direction: $(2\pi r \Delta r e_z)|_{z+\Delta z}$

work done by gravity: $(2\pi r \Delta r \Delta z \rho v_z g_z)$

$$(2\pi r e_r \Delta z)|_r - (2\pi r e_r \Delta z)|_{r+\Delta r} + (2\pi r \Delta r e_z)|_z - (2\pi r \Delta r e_z)|_{z+\Delta z} + (2\pi r \Delta r \Delta z \rho v_z g_z) = 0$$

$$\frac{(re_r)|_r}{\Delta r} - \frac{(re_r)|_{r+\Delta r}}{\Delta r} + r\frac{e_z|_z}{\Delta z} - r\frac{e_z|_{z+\Delta z}}{\Delta z} + r\rho g_z v_z = 0$$

$$-\frac{1}{r}\frac{(re_r)|_{r+\Delta r} - (re_r)|_r}{\Delta r} - \frac{(e_z)|_{z+\Delta z} - (e_z)|_z}{\Delta z} + \rho g_z v_z = 0$$

$$-\frac{1}{r}\frac{\partial}{\partial r}(re_r) - \frac{\partial}{\partial z}(e_z) + \rho g_z v_z = 0$$

using $\underline{e} = (\frac{1}{2}\rho v^2 + \rho\hat{H})\underline{v} + [\underline{\underline{\tau}}\cdot\underline{v}] + \underline{q}$ (9.8-6)

and $\hat{H} - \hat{H}_0 = \int_{T_0}^{T} \hat{C}_p dT + \int_{P_0}^{P}\left[\hat{V} - T\left(\frac{\partial \hat{V}}{\partial T}\right)_P\right] dP$

r component:

since $v_r = 0$, $(\frac{1}{2}\rho v^2 + \rho\hat{H})v_r = 0$

$e_r = \tau_{rz} v_z + q_r$

using Newton's Law & Fouriers Laws $\tau_{rz} = -\mu\frac{dv_z}{dr}$, $q_r = -k\frac{dT}{dr}$

$$e_r = \left(-\mu\frac{\partial v_z}{\partial r}\right)v_z - k\frac{\partial T}{\partial r}$$

z-component

$e_z = (\frac{1}{2}\rho v_z^2 + \rho\hat{H}) v_z + (\tau_{zx} v_x + \tau_{zy} v_y + \tau_{zz} v_z) + q_z$

Assumptions:

1. $v_x = v_y = 0$, $\tau_{zz} = -2\mu \frac{\partial v_z}{\partial z}$, $q_z = -k\frac{\partial T}{\partial z}$
2. $\hat{C}_p$ is not a function of T so $\int_{T_0}^{T} \hat{C}_p \, dT = \hat{C}_p (T - T_0)$
3. fluid has constant density so $\int_{P_0}^{P}\left[\hat{V} - T\left(\frac{\partial \hat{H}}{\partial T}\right)_p\right] d\bar{P} = \frac{P-P_0}{\rho}$

substituting

$$e_z = \frac{1}{2}\rho v_z^2 v_z + \rho \hat{H}_0 v_z + \hat{C}_p (T-T_0)\rho + (P-P_0)v_z - 2\mu \frac{\partial v_z}{\partial z} - k\frac{\partial T}{\partial z}$$

Substituting back into energy shell balance

$$-\frac{1}{r}\frac{\partial}{\partial r}\left(r\left[\left(-\mu\frac{\partial v_z}{\partial r}\right)v_z - k\frac{\partial T}{\partial r}\right]\right)$$

$$-\frac{\partial}{\partial z}\left(\frac{1}{2}\rho v_z^2 v_z + \rho \hat{H}_0 v_z + \hat{C}_p (T-T_0)\rho + (P-P_0)v_z - 2\mu \frac{\partial v_z}{\partial z} - k\frac{\partial T}{\partial z}\right]$$

$$+\rho g_z v_z = 0$$

since $v_z(r)$, not $v_z(z)$ $\frac{\partial v_z}{\partial z} = 0$

$$\rho \hat{C}_p v_z \frac{\partial T}{\partial z} = k\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \underbrace{\frac{\partial^2 T}{\partial z^2}}_{\text{neglected, small relative to transport by convection}}\right] + \underbrace{\mu\left(\frac{\partial v_z}{\partial r}\right)^2}_{\text{neglected (viscous heating)}} + v_z\overbrace{\left[-\frac{\partial P}{\partial z} + \mu\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \rho g\right]}^{=0 \text{ per Eq of motion}}$$

This simplifies (along w/ velocity distribution):

$$\rho \hat{C}_p v_{z,max}\left[1 - \left(\frac{r}{R}\right)^2\right]\frac{\partial T}{\partial z} = k\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right)\right]$$

BC1. at $r = 0$, T = finite (center of pipe is finite temp)

BC2: at $r = R$, $k\frac{\partial T}{\partial r} = q_0$ (constant flux in from wall)

BC3: at $z = 0$, $T = T_1$ (fluid enters segment at given temp)

CHE 312 HW04 (continued)

1. (continued)

Dimensionless Variables

let $\xi = \frac{r}{R}$, $\Theta = \frac{T-T_1}{q_0 R/k}$, $\zeta = \frac{z}{\rho \hat{C}_p v_{zmax} R^2/k}$

$\frac{\partial \xi}{\partial r} = \frac{1}{R}$ $\frac{\partial \zeta}{\partial z} = \frac{1}{\rho \hat{C}_p v_{zmax} R^2/k}$

$$\rho \hat{C}_p v_{zmax}\left[1-(\xi)^2\right]\frac{\partial \Theta}{\partial \zeta}\frac{1}{\rho \hat{C}_p v_{zmax} R^2/k} = k\left[\left(\frac{1}{r}\frac{R^3 \partial}{\partial \xi}\left(\frac{r}{R}\frac{\partial \Theta}{\partial \xi}\right)\right)\right]$$

$$(1-\xi^2)\frac{\partial \Theta}{\partial \zeta} = \frac{1}{\xi}\frac{\partial}{\partial \xi}\left(\xi \frac{\partial \Theta}{\partial \xi}\right)$$

BC1: $\xi = 0$, Θ = finite BC3: $\zeta = 0$, $\Theta = 0$

BC2: $\xi = 1$ $\frac{\partial \Theta}{\partial \xi} = 1$

using insight that temperature profile will stabilize in shape after large ζ, we approximate Θ as follows

$\Theta(\xi,\zeta) = C_0\zeta + \psi(\xi)$, where ψ is a function of ξ

Since BC3 is no longer workable with asymptotic solution, a new BC is generated using macroscopic energy balance that heat added leads to temp gained.

$$2\pi R z q_0 = \int_0^{2\pi}\int_0^R \rho \hat{C}_p (T-T_1) v_z r\,dr\,d\theta$$

$$\frac{\cancel{2\pi}\cancel{R} z q_0}{\rho \hat{C}_p} = \int_0^{\cancel{2\pi}}\int_0^R \cancel{q_0}\frac{\cancel{R}}{k}\Theta v_{zmax}\left[1-\xi^2\right] r\,dr\,d\cancel{\theta}$$

$$\zeta = \int_0^1 \Theta(1-\xi^2)\,d\xi$$

Inserting $\Theta = C_0\zeta + \psi(\xi)$ into PDE

$(1-\xi^2)C_0 = \frac{1}{\xi}\frac{\partial}{\partial \xi}\left(\xi \frac{\partial \psi}{\partial \xi}\right)$

$\xi \frac{\partial \psi}{\partial \xi} = C_0\left(\frac{1}{2}\xi^2 - \frac{1}{4}\xi^4\right) + C_1 \Rightarrow \frac{\partial \psi}{\partial \xi} = C_0\left(\frac{1}{2}\xi - \frac{1}{4}\xi^3\right) + \frac{C_1}{\xi}$

$\psi = C_0\left(\frac{1}{4}\xi^2 - \frac{1}{16}\xi^4\right) + C_1 \ln \xi + C_2$

$\Theta = C_0\zeta + C_0\left(\frac{1}{4}\xi^2 - \frac{1}{16}\xi^4\right) + C_1 \ln \xi + C_2$

Applying BCS.

BC1: since $\ln(0)$ not defined, $C_1 = 0$

$\frac{\partial \Theta}{\partial \xi} = C_0\left(\frac{1}{2}\xi - \frac{1}{4}\xi^3\right)$

BC2:

$1 = c_0\left(\frac{1}{2} - \frac{1}{4}\right) + \quad \Rightarrow \quad c_0 = 4 \quad c_2$

BC3: at $\xi = 0$

$0 = \int_0^1 \left(\xi^2 - \frac{1}{4}\xi^4 + c_2\right)(1 - \xi^2)\, d\xi \Rightarrow 0 = \int_0^1 \xi^2 - \xi^4 - \frac{1}{4}\xi^4 + \frac{1}{4}\xi^6 + c_2 -$

$0 = \frac{1}{3} - \frac{1}{5} - \frac{1}{20} + \frac{1}{28} - \frac{1}{3}c_2 + c_2 \quad c_2 = -7/24$

$$\boxed{\Theta = 4\zeta + \xi^2 - \frac{1}{4}\xi^4 - \frac{7}{24}}$$

For temp average at a given z, we integrate Θ over the cross-sectional area

$$\frac{T - T_1}{q_0 R/k} = 4\frac{z}{\rho \hat{C}_p v_{max} R^2/k} + \left(\frac{r}{R}\right)^2 - \frac{1}{4}\left(\frac{r}{R}\right)^4 - \frac{7}{24}$$

$$T = \frac{q_0 R}{k}\left[\frac{4z}{\rho \hat{C}_p v_{max} R^2/k} + \left(\frac{r}{R}\right)^2 - \frac{1}{4}\left(\frac{r}{R}\right)^4 - \frac{7}{24}\right] + T_1$$

$$\langle T \rangle = \frac{\int_0^{2\pi}\int_0^R T \cdot r\, dr\, d\theta}{\int_0^{2\pi}\int_0^R r\, dr\, d\theta} \qquad \pi R^2$$

$$\int_0^{2\pi}\int_0^R \frac{q_0 R}{k}\left[4\zeta r + \frac{1}{R^2} r^3 - \frac{1}{4R^4} r^5 - \frac{7}{24} r\right] + T_1 r\, dr\, d\theta$$

$$\frac{q_0 R}{k}\int_0^{2\pi}\left[2\zeta r^2 + \frac{1}{R^2 4} r^4 - \frac{1}{24R^4} r^6 - \frac{7}{48} r^2 + \frac{k}{q_0 R 2} T_1 r^2 \Big|_0^R\right] d\theta$$

$$\left(\left(2\zeta R^2 + \frac{1}{4}R^2 - \frac{1}{24}R^2 - \frac{7}{48}R^2\right)\frac{q_0 R}{k} + T_1 R\right) 2\pi$$

divide by area (πR^2)

$$\boxed{\langle T \rangle = \left(4\zeta + \frac{1}{8}\right)\frac{q_0 R}{k} + T_1}$$

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CHE 312 HW04 (continued)

1. (continued x2)

For the cup mixing temperature:

$$T_b = \frac{\langle v_z T\rangle}{\langle v_z\rangle} = \frac{\int_0^{2\pi}\int_0^R v_z \cdot T \cdot r\,dr\,d\theta}{\int_0^{2\pi}\int_0^R v_z\,r\,dr\,d\theta} \qquad v_z = v_{zmax}(1-\xi^2)$$

$$\int_0^{2\pi}\int_0^{R} v_{zmax}\left[1-\left(\frac{r}{R}\right)^2\right]\left(\frac{q_0 R}{k}\left[4\xi + \left(\frac{r}{R}\right)^2 - \frac{1}{4}\left(\frac{r}{R}\right)^4 - \frac{7}{24}\right] + T_1\right) r\,dr\,d\theta \quad dr\,d\theta$$

$$v_{zmax}\iint\left(\frac{q_0 R}{K}\left[4\xi r + \frac{1}{R^2}r^3 - \frac{1}{4R^4}r^5 - \frac{7}{24}r\right] + T_1 r\right) - \left(\frac{q_0 R}{K}\left[4\xi\frac{1}{R^2}r^3 + \frac{1}{R^4}r^5 - \frac{1}{4R^6}r^7 - \frac{7}{24R^2}r^3\right] + \frac{T_1}{R^2}r^3\right)$$

$$v_{zmax}\int\left(\frac{q_0 R}{K}\left[2\xi R^2 + \frac{1}{4}R^2 - \frac{1}{24}R^2 - \frac{7}{48}R^2\right] + \frac{T_1}{2}R^2\right) - \left(\frac{q_0 R}{k}\left[4\xi\frac{1}{4}R^2 + \frac{1}{6}R^2 - \frac{1}{32}R^2 - \frac{7}{96}R^2\right] + \frac{T_1}{4}R^2\right)$$

$$\frac{\cancel{2\pi v_{zmax}}\, q_0 R}{k}\left[\left(\frac{4}{2}\xi + \frac{3}{48}\right)R^2 + \frac{T_1 k}{2q_0 R}R^2 - \left(\frac{4}{4}\xi + \frac{1}{16}\right)R^2 + \frac{T_1 k}{4q_0 R}R^2\right]$$

$$\iint v_{zmax}\left(1-\frac{1}{R^2}r^2\right) r\,dr\,d\theta$$

$$v_{zmax}\cdot 2\pi\int r - \frac{1}{R^2}r^3\,dr \qquad \left(\cancel{\frac{1}{2}R^2 - \frac{1}{4}R^2}\right)\left(4\xi + T_1\frac{k}{q_0 R}\right)\frac{q_0 R}{k}$$

$$\cancel{v_{zmax}\,2\pi}\left[\cancel{\frac{1}{2}R^2 - \frac{1}{4}R^2}\right]$$

$$\boxed{T_b = T_1 + (4\xi)\frac{q_0 R}{k}}$$

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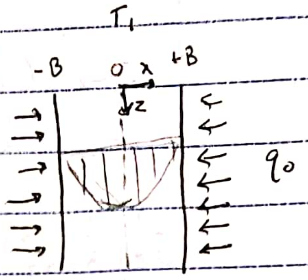


2. [10.B-7]

a. Velocity distribution

Equation of continuity

$$\cancel{\frac{\partial \rho}{\partial t}} + \frac{\partial}{\partial x}(\cancel{\rho v_x}) + \frac{\partial}{\partial y}(\rho v_y) + \cancel{\frac{\partial}{\partial z}(\rho v_z)} = 0$$

const ρ $\quad v_x = 0 \quad v_y = 0 \quad v_z \neq v_z(z)$

$v_z(x)$

$P(z)$

$0 = 0$

Equation of Motion

x: $\quad 0 = 0$

y: $\quad 0 = 0$

$$z: \rho\left(\cancel{\frac{\partial v_z}{\partial t}} + \cancel{v_x \frac{\partial v_x}{\partial x}} + \cancel{v_y \frac{\partial v_y}{\partial y}} + \cancel{v_z \frac{\partial v_z}{\partial z}}\right) = -\frac{\partial p}{\partial z} + \mu\left[\frac{\partial^2 v_z}{\partial x^2} + \cancel{\frac{\partial^2 v_z}{\partial y^2}} + \cancel{\frac{\partial^2 v_z}{\partial z^2}}\right] + \rho g_z$$

const ρ $\quad v_x = 0 \quad v_y = 0 \quad v_z \neq v_z(z) \quad v_z \neq v_z(y) \quad v_z \neq v_z(z)$

$$0 = -\frac{\partial p}{\partial z} + \mu \frac{\partial^2 v_z}{\partial x^2} + \rho g_z \qquad \text{let } -\frac{\partial \mathcal{P}}{\partial z} = -\frac{\partial p}{\partial z} + \rho g_z$$

$$\frac{\partial \mathcal{P}}{\partial z} = \mu \frac{\partial^2 v_z}{\partial z^2} = c_0$$

$\mathcal{P} = c_0 z + c_1 \quad$ BC1: $\mathcal{P}(0) = \mathcal{P}_0$, $\quad$ BC2: $\mathcal{P}(L) = \mathcal{P}_L \quad c_1 = \mathcal{P}$

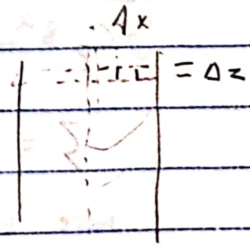
$$\mathcal{P} = \frac{\mathcal{P}_L - \mathcal{P}_0}{L} \cdot z + \mathcal{P}_0$$

$$\frac{\partial^2 v_z}{\partial x^2} = \frac{1}{\mu}\frac{\mathcal{P}_L - \mathcal{P}_0}{L}$$

$$\frac{\partial v_z}{\partial x} = \frac{(\mathcal{P}_L - \mathcal{P}_0)}{\mu L} \cdot x + C_2 \qquad \text{BC} = \frac{\partial v_z(0)}{\partial x} = 0, \quad C_2 = 0$$

$$v_z = \frac{\mathcal{P}_L - \mathcal{P}_0}{2\mu L} x^2 + C_3 \qquad \text{BC: } v_z(B) = 0 \qquad C_3 = -\frac{\mathcal{P}_L - \mathcal{P}_0}{2\mu L} B^2$$

$$v_z = \frac{(\mathcal{P}_0 - \mathcal{P}_L) B^2}{2\mu L}\left[1 - \left(\frac{x}{B}\right)^2\right] = v_{z\max}\left[1 - \left(\frac{x}{B}\right)^2\right]$$



Energy Shell Balance

energy in from z: $(W\Delta x \cdot e_z)|_z$

energy out from z: $(W\Delta x \cdot e_z)|_{z+\Delta z}$

energy in from x: $(W \cdot \Delta z \cdot e_x)|_x$

energy out from x: $(W \cdot \Delta z\, e_x)|_{x+\Delta x}$

work done: $\quad (W \cdot \Delta x \Delta z\, \rho v_z g_z)$

$$\Delta x W(e_z)|_z - \Delta x W(e_z)|_{z+\Delta z} + \Delta z W(e_x)|_x - \Delta z W(e_x)|_{x+\Delta x} + \Delta x \Delta z W \rho v_z g_z = 0$$

$$\frac{e_z|_{z+\Delta z} - e_z|_z}{\Delta z} + \frac{(e_x)|_{x+\Delta x} - (e_x)|_x}{\Delta x} - \rho v_z g_z = 0$$

$$\frac{\partial e_z}{\cdot} + \frac{\partial e_x}{\cdot} - \ldots$$

$$e_x = \cancel{\left(\tfrac{1}{2}\rho v_x^2 + \rho \hat{H}\right)v_x} + [\tau \cdot v]_x + q_x = -v_z\left(\mu \frac{\partial v_z}{\partial x}\right) - k\frac{\partial T}{\partial x}$$

$$e_z = \left(\tfrac{1}{2}\rho v_z^2\right)v_z + \rho \hat{H}^\circ v_z + (p - p_0)v_z + \rho \hat{C}_p (T - T_0)v_z - \left(2\mu \cancel{\frac{\partial v_z}{\partial z}}\right)v_z - \cancel{k\frac{\partial T}{\partial z}}$$

$$\frac{\partial}{\partial z}\left[\cancel{\left(\tfrac{1}{2}\rho v_z^2\right)v_z} + \cancel{\rho \hat{H}^\circ v_z} + (p - p_0)v_z + \rho \hat{C}_p (T - T_0)v_z\right] \Rightarrow \frac{\partial p}{\partial z}v_z + \rho \hat{C}_p \frac{\partial T}{\partial z}v_z \quad \text{viscous}$$

$$+\frac{\partial}{\partial x}\left[\left(-\mu \frac{\partial v_z}{\partial x}\right)v_z - k\frac{\partial T}{\partial x}\right] - \rho v_z g = 0 \Rightarrow -\mu v_z \frac{\partial^2 v_z}{\partial x^2} + \cancel{\left(-\mu \frac{\partial v_z}{\partial x}\cdot\frac{\partial v_z}{\partial x}\right)} - k\frac{\partial^2 T}{\partial x^2}$$

$$\frac{\partial p}{\partial z}v_z + \rho \hat{C}_p \frac{\partial T}{\partial z}v_z + \left(-\mu v_z \frac{\partial^2 v_z}{\partial x^2}\right) - k\frac{\partial^2 T}{\partial x^2} - \rho v_z g = 0$$

$$v_z\underbrace{\left[\frac{\partial p}{\partial z} - \mu \frac{\partial^2 v_z}{\partial x^2} - \rho g\right]}_{=0 \text{ per Eq of Motion}} + \rho \hat{C}_p \frac{\partial T}{\partial z}v_z = k\frac{\partial^2 T}{\partial x^2} \Rightarrow \boxed{\rho \hat{C}_p \frac{\partial T}{\partial z}v_{max}\left[1 - \left(\frac{x}{B}\right)^2\right] = k\frac{\partial^2 T}{\partial x^2}}$$

b. $\Theta = \dfrac{T - T_1}{q_0 B/k}$ $\quad \sigma = \dfrac{x}{B}$ $\quad \zeta = \dfrac{kz}{\rho \hat{C}_p v_{max} B^2}$ $\quad \dfrac{\partial \sigma}{\partial x} = \dfrac{1}{B}$ $\quad \dfrac{\partial \zeta}{\partial z} = \dfrac{k}{\rho \hat{C}_p v_{max} B^2}$

$$\rho \hat{C}_p \frac{\partial \Theta}{\partial z}v_{max}[1 - \sigma^2] = k\frac{\partial^2 \Theta}{\partial x^2}$$

$$\rho \hat{C}_p \frac{\partial \Theta}{\partial \zeta}\frac{k}{\rho \hat{C}_p v_{max} B^2}\cdot v_{max}[1 - \sigma^2] = \frac{k}{B^2}\frac{\partial^2 \Theta}{\partial \sigma^2}$$

$$\boxed{\frac{\partial \Theta}{\partial \zeta}[1 - \sigma^2] = \frac{\partial^2 \Theta}{\partial \sigma^2} \qquad \text{BC1: at } \sigma = -1,\ \frac{\partial \Theta}{\partial \sigma} = -1 \qquad \text{BC2: at } \sigma = 1,\ \frac{\partial \Theta}{\partial \sigma} = 1}$$

c. To find asymptotic solution, propose at large z:

$$Wz\, q_0 = \int_0^W \int_{-B}^{B} \rho \hat{C}_p (T - T_1) v_z \, dx\, dy$$

$$\frac{kz}{\rho C_p B} = \int_{-B}^{B} \frac{k(T - T_1)}{q_0 B} v_{z\,max}\left[1 - \left(\frac{x}{B}\right)^2\right] dx$$

$$\frac{kz}{\rho C_p B^2 v_{max}} = \int_{-1}^{1} \frac{k(T - T_1)}{q_0 B}\left[1 - \left(\frac{x}{B}\right)^2\right] d\left(\frac{x}{B}\right)$$

$$\zeta = \int_{-1}^{1} \Theta[1 - \sigma^2]\, d\sigma$$

where $\Theta = C_0 \zeta + \psi(\sigma)$

$$C_0[1 - \sigma^2] = \frac{\partial^2 \psi}{\partial \sigma^2}$$

$$\frac{\partial \psi}{\partial \sigma} = C_0\left[\sigma - \tfrac{1}{3}\sigma^3\right] + C_1$$

$$\psi = C_0\left[\tfrac{1}{2}\sigma^2 - \tfrac{1}{12}\sigma^4\right] + C_1 \sigma + C_2$$

$$\Theta = C_0 \zeta + C_0\left[\tfrac{1}{2}\sigma^2 - \tfrac{1}{12}\sigma^4\right] + C_1 \sigma + C_2 \qquad \frac{\partial \Theta}{\partial \sigma} = C_0\left[\sigma - \tfrac{1}{3}\sigma^3\right] + C_1$$

using BC1:

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CHE 312 HW04 (continued)

2. (continued)

BC1: $-1 = C_0[-1+\frac{1}{3}] + C_1 \Rightarrow -\frac{2}{3}C_0 = -1 - C_1$ } $C_1 = 0$

BC2: $1 = C_0[1-\frac{1}{3}] + C_1 \Rightarrow \frac{2}{3}C_0 = 1 - C_1$ } $C_0 = \frac{3}{2}$

BC3: $\zeta = \int_{-1}^{1} (1-\sigma^2)\left(\frac{3}{2}\zeta + \frac{3}{2}\left[\frac{1}{2}\sigma^2 - \frac{1}{12}\sigma^4\right] + C_2\right) d\sigma$ at $\zeta = 0$;

$0 = \int_{-1}^{1} (1-\sigma^2)\left(\frac{3}{4}\sigma^2 - \frac{1}{8}\sigma^4 + C_2\right) d\sigma$

$0 = \int_{-1}^{1} \left(\frac{3}{4}\sigma^2 - \frac{1}{8}\sigma^4 + C_2 - \frac{3}{4}\sigma^4 + \frac{1}{8}\sigma^6 - C_2\sigma^2\right) d\sigma$

$0 = \frac{3}{12}\sigma^3 - \frac{1}{40}\sigma^5 + C_2\sigma - \frac{3}{20}\sigma^5 + \frac{1}{56}\sigma^7 - C_2\frac{1}{3}\sigma^3 \Big|_{-1}^{1}$

$0 = \frac{1}{2} - \frac{1}{20} + 2C_2 - \frac{3}{10} + \frac{1}{28} - \frac{2}{3}C_2$ $\quad C_2 = -39/280$

$$\boxed{\Theta = \frac{3}{2}\zeta + \frac{3}{2}\sigma^2 - \frac{1}{8}\sigma^4 - \frac{39}{280}}$$

Homework #4 10/28

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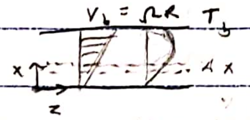
CHE 312 HW04 (continued)

2. Using the simplification that, at large R and small b, the system can be analogous to parallel plate movements.

Shell Balance



energy in from x: $WLe_x|_x$

energy out from x: $WLe_x|_{x+\Delta x}$

$v(x) = V_b\left(\frac{x}{b}\right)$

$$WLe_x|_x - WLe_x|_{x+\Delta x} = 0$$

$$\frac{de_x}{dx} = 0 \Rightarrow e_x = C_1$$

using $e_x = \cancel{(\frac{1}{2}\rho v^2 + \rho \hat{H}) v_x} + (\cancel{\tau_{xx}v_x} + \cancel{\tau_{xy}v_y} + \tau_{xz}v_z) + q_x$

$$e_x = -\mu \frac{\partial v_z}{\partial x} v_z - k\frac{dT}{dx} = C_1 \quad \text{using } v_z = V_b \frac{x}{b}, \quad \frac{dv_z}{dx} = V_b \frac{1}{b}$$

$$-\mu x\left(\frac{V_b}{b}\right)^2 - k\frac{dT}{dx} = C_1 \Rightarrow \frac{dT}{dx} = \frac{-1}{k}\left[C_1 + \mu x\left(\frac{V_b}{b}\right)^2\right]$$

$$T = -\frac{1}{k}C_1 x - \frac{1}{2k}\mu x^2\left(\frac{V_b}{b}\right)^2 + C_2$$

using BC1: at $x = 0$, $T = T_0 \Rightarrow C_2 = T_0$

using BC2: at $x = b$, $T = T_b$

$$T_b = -\frac{1}{k}C_1 b - \frac{1}{2k}\mu V_b^2 + T_0$$

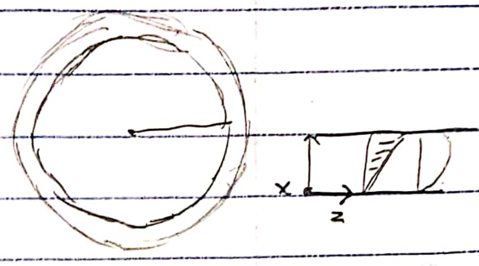
$$k(T_b - T_0) = -C_1 b - \frac{1}{2}\mu V_b^2 \qquad C_1 = -\frac{1}{2b}\mu V_b^2 + \frac{k}{b}(T_b - T_0)$$

$$T = -\frac{1}{k}\left[\frac{1}{2b}\mu V_b^2 + \frac{k}{b}(T_b - T_0)\right]x + \frac{1}{2k}\mu V_b^2 \frac{x^2}{b^2} + T_0$$

$$\frac{T - T_0}{T_b - T_0} = \frac{+1}{k}\cdot\frac{1}{2b}\frac{\mu V_b^2}{(T_b - T_0)}x + \frac{1}{k}\frac{x}{b} - \frac{1}{2k}\mu\frac{V_b^2}{(T_b - T_0)}\frac{x^2}{b^2} \quad \text{using } Br = \frac{\mu V_b^2}{k(T_b - T_0)}$$

$$\boxed{\frac{T - T_0}{T_b - T_0} = \frac{1}{2}Br\frac{x}{b}\left[1 - \frac{x}{b}\right] + \frac{x}{b}}$$

CHE 312 HW04 (continued)



4. [10.B-2]

$v_z = v_b\left(\frac{x}{b}\right)$ $v_b = \Omega R$

Shell balance

energy in : $WL\, e_x|_x$ BC1 at $x=0$, $T=T_0$

energy out : $WL\, e_x|_{x+\Delta x}$ BC2 at $x=b$, $q_x=0$

$WL e_x|_x - WL e_x|_{x+\Delta x} = 0$

$\frac{\partial e_x}{\partial x} = 0$, so $e_x = C_1$

$e_x = (\cancel{\tfrac{1}{2}\rho v_x^2 + \rho H})\, v_x^{\,0} + [\tau \cdot v_x] + q_x = \cancel{\tau_{xx} v_x}^{\,0} + \cancel{\tau_{xy} v_y}^{\,0} + \tau_{xz} v_z - k\frac{\partial T}{\partial x}$

$e_x = -\mu \frac{\partial v_z}{\partial x} v_z - k\frac{\partial T}{\partial x} = C_1$ where $\frac{\partial v_z}{\partial x} = \frac{v_b}{b}$

$-\mu\, v_b\left(\frac{x}{b}\right)\frac{v_b}{b} - k\frac{\partial T}{\partial x} = C_1 \Rightarrow -\mu x\left(\frac{v_b}{b}\right)^2 - k\frac{\partial T}{\partial x} = C_1$; BC2 $\Rightarrow$ $C_1 = -\mu b\left(\frac{v_b}{b}\right)^2$

$\frac{\partial T}{\partial x} = -\frac{1}{k}C_1 - \frac{\mu}{k}x\left(\frac{v_b}{b}\right)^2$

$T = -\frac{1}{k}C_1 x - \frac{\mu}{2k}x^2\left(\frac{v_b}{b}\right)^2 + C_2 \Rightarrow T = \frac{\mu v_b^2}{kb}\cdot x - \frac{\mu v_b^2}{2kb^2}x^2 + C_2$

BC1: $T_0 = C_2$

$T = \left(\frac{\mu v_b^2}{k}\right)\left[\frac{x}{b} - \frac{1}{2b^2}x^2\right] + T_0$

$$\boxed{\frac{T - T_0}{\mu v_b^2/k} = \left(\frac{x}{b}\right) - \frac{1}{2}\left(\frac{x}{b}\right)^2}$$

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